

Statistical Learning and Machine Learning

Exercise Session (Week 11)

SOLUTIONS

BOOK EXERCISES

Problem 5.28

See <https://www.microsoft.com/en-us/research/wp-content/uploads/2016/05/prml-web-sol-2009-09-08.pdf#page=52>

CONVOLUTIONAL NEURON

- 1) The shape of the output after applying the convolutional neuron $\mathbf{F}$ to $\mathbf{X}_1$ will be $((3-3)/1+1) \times ((4-3)/1+1) = 1 \times 2$. The convolution is given by $(\mathbf{X}_1 * \mathbf{F})[n, m] = \sum_{j=0}^2 \sum_{i=0}^2 \mathbf{X}_1[n+i, m+j] \cdot \mathbf{F}[2-i, 2-j]$ for $n = 0$ and $m = 0, 1$.

$$\begin{aligned} (\mathbf{X}_1 * \mathbf{F})[0, 0] &= \sum_{j=0}^2 \sum_{i=0}^2 \mathbf{X}_1[0+i, 0+j] \cdot \mathbf{F}[2-i, 2-j] \\ &= 1 \cdot 1 + 0 \cdot 2 - 1 \cdot 3 + 2 \cdot 4 + 0 \cdot 5 - 2 \cdot 7 + 1 \cdot 9 + 0 \cdot 10 - 1 \cdot 11 \\ &= 1 + 0 - 3 + 8 + 0 - 14 + 9 + 0 - 11 \\ &= -10 \end{aligned}$$

and

$$\begin{aligned} (\mathbf{X}_1 * \mathbf{F})[0, 1] &= \sum_{j=0}^2 \sum_{i=0}^2 \mathbf{X}_1[0+i, 1+j] \cdot \mathbf{F}[2-i, 2-j] \\ &= 1 \cdot 2 + 0 \cdot 3 - 1 \cdot 4 + 2 \cdot 5 + 0 \cdot 7 - 2 \cdot 8 + 1 \cdot 10 + 0 \cdot 11 - 1 \cdot 12 \\ &= 2 + 0 - 4 + 10 + 0 - 16 + 10 + 0 - 12 \\ &= -10 \end{aligned}$$

Hence, $(\mathbf{X}_1 * \mathbf{F}) = \begin{bmatrix} -10 & -10 \end{bmatrix}$

- 2) Zero-padding the input matrix by 1, corresponds to increasing its size by 2 in both directions. Let $\mathbf{X}_z$ be the $\mathbf{X}_1$ zero-padded by 1:

$$\mathbf{X}_z = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & 4 & 0 \\ 0 & 4 & 5 & 7 & 8 & 0 \\ 0 & 9 & 10 & 11 & 12 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The output shape after applying $\mathbf{F}$ to $\mathbf{X}_z$ becomes $((5-3)/1+1) \times ((6-3)/1+1) = 3 \times 4$. The convolution is given by $(\mathbf{X}_z * \mathbf{F})[n, m] = \sum_{j=0}^2 \sum_{i=0}^2 \mathbf{X}_z[n+i, m+j] \cdot \mathbf{F}[2-i, 2-j]$ for $n = 0, 1, 2$ and $m = 0, 1, 2, 3$. Hence, result of applying $\mathbf{F}$ to $\mathbf{X}_z$ is:

$$(\mathbf{X}_z * \mathbf{F}) = \begin{bmatrix} -9 & -7 & -7 & -9 \\ -22 & -10 & -10 & 28 \\ -25 & -7 & -7 & 29 \end{bmatrix}$$

- 3) The output shape after applying $\mathbf{F}$ to $\mathbf{X}_2$ becomes $((224-3)/1+1) \times ((224-3)/1+1) = 222 \times 222$.
 4) The output shape after applying $\mathbf{F}$ to $\mathbf{X}_2$ with a stride of 2 becomes $(\lfloor (224-3)/2 \rfloor + 1) \times (\lfloor (224-3)/2 \rfloor + 1) = 111 \times 111$.

Note: Since we are given no padding scheme, we have to round down when dividing by the stride. When the kernel reaches the opposite side of the input matrix, the 3rd column will not be supported by the input matrix. Hence, we have to round down, as there is only room for 111 supported convolutions in either direction.

BACKPROPAGATION

- 1) Augmenting the input vector $\mathbf{x}$ and the output of the hidden layer $k = 1$ with a 1 to account for the bias, the activation's $\mathbf{a}_k$, output of the activation functions $\mathbf{z}_k$, and the output becomes:

$$\mathbf{a}_1 = \mathbf{w}_1^T \begin{bmatrix} 1 \\ \mathbf{x} \end{bmatrix} = \begin{bmatrix} 0.040 \\ -1.180 \\ -0.740 \end{bmatrix} \quad \mathbf{z}_1 = h(\mathbf{a}_1) = \begin{bmatrix} 0.510 \\ 0.235 \\ 0.323 \end{bmatrix}$$

$$\mathbf{a}_2 = \mathbf{w}_2^T \begin{bmatrix} 1 \\ \mathbf{z}_1 \end{bmatrix} = \begin{bmatrix} -0.746 \\ 0.226 \end{bmatrix} \quad \mathbf{y}(\mathbf{x}, \{\mathbf{w}_1, \mathbf{w}_2\}) = \mathbf{z}_2 = h(\mathbf{a}_2) = \begin{bmatrix} 0.322 \\ 0.556 \end{bmatrix}$$

- 2) The cross-entropy error for multiple separate binary classification output is given by equation (5.23) in the book, and can be shortend to:

$$E(\mathbf{x}_1, \{\mathbf{w}_1, \mathbf{w}_2\}) = - \sum_{c=1}^2 t_c \ln(y_c) + (1 - t_c) \ln(1 - y_c)$$

where c is the index of the output of the network. As $t_c = 0$ for both outputs, the cross-entropy error becomes:

$$E(\mathbf{x}_1, \{\mathbf{w}_1, \mathbf{w}_2\}) = - \sum_{c=1}^2 \ln(1 - y_c) = - \ln(1 - 0.322) - \ln(1 - 0.556) = 0.388 + 0.812 = 1.20$$

- 3) Using backpropagation, the derivative of the error w.r.t. the weights can be found using the chain rule for partial derivatives. From exercise 5.6 in the book, the cross-entropy error w.r.t. the activation of the last layer (given a sigmoid activation function) is:

$$\frac{\partial E}{\partial \mathbf{a}_2} = \mathbf{y} - \mathbf{t} = \begin{bmatrix} 0.322 \\ 0.556 \end{bmatrix}$$

Using the chain rule (equation 5.53 in the book), the derivative of the error w.r.t. to the weights of the second layer become:

$$\begin{bmatrix} \frac{\partial E}{\partial w_{0,1}^{(2)}} & \frac{\partial E}{\partial w_{0,2}^{(2)}} \\ \frac{\partial E}{\partial w_{1,1}^{(2)}} & \frac{\partial E}{\partial w_{1,2}^{(2)}} \\ \frac{\partial E}{\partial w_{2,1}^{(2)}} & \frac{\partial E}{\partial w_{2,2}^{(2)}} \\ \frac{\partial E}{\partial w_{3,1}^{(2)}} & \frac{\partial E}{\partial w_{3,2}^{(2)}} \end{bmatrix} = \frac{\partial E}{\partial \mathbf{w}_2} = \frac{\partial \mathbf{a}_2}{\partial \mathbf{w}_2} \frac{\partial E}{\partial \mathbf{a}_2}^T = \begin{bmatrix} 1 \\ \mathbf{z}_1 \end{bmatrix} \frac{\partial E}{\partial \mathbf{a}_2}^T = \begin{bmatrix} 0.322 & 0.556 \\ 0.164 & 0.284 \\ 0.076 & 0.131 \\ 0.104 & 0.180 \end{bmatrix}$$

Hence, $\frac{\partial E}{\partial w_{0,2}^{(2)}} = 0.556$.

To determine the error w.r.t. the weights in the first layer, the error needs to be backpropagated to the activations of the hidden layer $k = 1$ using equation 5.56 in the book:

$$\frac{\partial E}{\partial \mathbf{a}_1} = \frac{\partial \mathbf{z}_1}{\partial \mathbf{a}_1} \frac{\partial \mathbf{a}_2}{\partial \mathbf{z}_1} \frac{\partial E}{\partial \mathbf{a}_2}$$

where $\frac{\partial \mathbf{z}_1}{\partial \mathbf{a}_1} = h(\mathbf{a}_1) \odot (1 - h(\mathbf{a}_1)) = \mathbf{z}_1 \odot (1 - \mathbf{z}_1)$ is the derivative of the activation function (eq. 4.88 in the book), and $\frac{\partial \mathbf{a}_2}{\partial \mathbf{z}_1} = \mathbf{w}_2$ is the derivative of the second layer. Hence, the derivative of the error w.r.t. the activations of the hidden layer becomes:

$$\frac{\partial E}{\partial \mathbf{a}_1} = (\mathbf{z}_1 \odot (1 - \mathbf{z}_1)) \odot \mathbf{w}_2 \frac{\partial E}{\partial \mathbf{a}_2}$$

$$\frac{\partial E}{\partial \mathbf{a}_1} = \begin{bmatrix} 0.510(1 - 0.510) \\ 0.235(1 - 0.235) \\ 0.323(1 - 0.313) \end{bmatrix} \odot \begin{bmatrix} 0.7 & 0.5 \\ -1.0 & 0.3 \\ 0.1 & 0.0 \end{bmatrix} \begin{bmatrix} 0.322 \\ 0.556 \end{bmatrix} = \begin{bmatrix} 0.126 \\ -0.028 \\ 0.007 \end{bmatrix}$$

Note: The bias from the hidden layer has been discarded, when backpropagating the error to the activations of the hidden layer. Similar to the hidden layer, the derivative of the error w.r.t. the weights of the first layer can be found using equation 5.53 in the book:

$$\begin{bmatrix} \frac{\partial E}{\partial w_{0,1}^{(1)}} & \frac{\partial E}{\partial w_{0,2}^{(1)}} & \frac{\partial E}{\partial w_{0,3}^{(1)}} \\ \frac{\partial E}{\partial w_{1,1}^{(1)}} & \frac{\partial E}{\partial w_{1,2}^{(1)}} & \frac{\partial E}{\partial w_{1,3}^{(1)}} \\ \frac{\partial E}{\partial w_{2,1}^{(1)}} & \frac{\partial E}{\partial w_{2,2}^{(1)}} & \frac{\partial E}{\partial w_{2,3}^{(1)}} \end{bmatrix} = \frac{\partial E}{\partial \mathbf{w}_1} = \frac{\partial \mathbf{a}_1}{\partial \mathbf{w}_1} \frac{\partial E}{\partial \mathbf{a}_1}^T = \begin{bmatrix} 1 \\ \mathbf{x} \end{bmatrix} \frac{\partial E}{\partial \mathbf{a}_1}^T = \begin{bmatrix} 0.126 & -0.029 & 0.007 \\ -0.050 & 0.011 & -0.003 \\ 0.000 & 0.000 & 0.00 \end{bmatrix}$$

Therefore, $\frac{\partial E}{\partial w_{1,1}^{(1)}} = -0.050$.